

## Credit Screening Analysis & Risk Engine

### The LOGISTIC Procedure

| Model Information         |                         |
|---------------------------|-------------------------|
| Data Set                  | WORK.UCI_TRANSFORMED    |
| Response Variable         | Target_Binary           |
| Number of Response Levels | 2                       |
| Model                     | binary logit            |
| Optimization Technique    | Fisher's scoring        |
| Likelihood Penalty        | Firth's bias correction |

|                             |    |
|-----------------------------|----|
| Number of Observations Read | 15 |
| Number of Observations Used | 15 |

| Response Profile |               |                 |
|------------------|---------------|-----------------|
| Ordered Value    | Target_Binary | Total Frequency |
| 1                | 0             | 5               |
| 2                | 1             | 10              |

Probability modeled is Target\_Binary=1.

| Intercept-Only Model Convergence Status       |
|-----------------------------------------------|
| Convergence criterion (GCONV=1E-8) satisfied. |

| Model Convergence Status                      |
|-----------------------------------------------|
| Convergence criterion (GCONV=1E-8) satisfied. |

| Model Fit Statistics |                |                          |
|----------------------|----------------|--------------------------|
| Criterion            | Intercept Only | Intercept and Covariates |
| AIC                  | 8.610          | 6.521                    |
| SC                   | 9.318          | 9.353                    |
| -2 Log L             | 6.610          | -1.479                   |

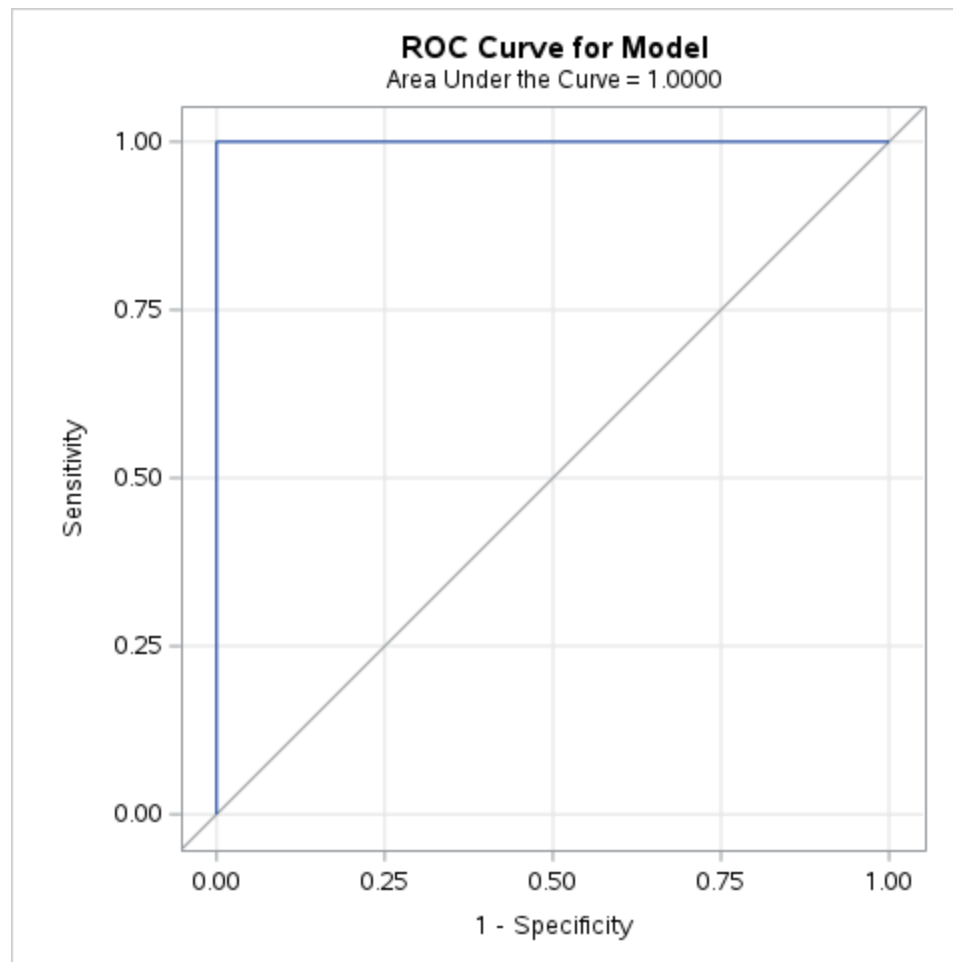
| Testing Global Null Hypothesis: BETA=0 |            |    |            |
|----------------------------------------|------------|----|------------|
| Test                                   | Chi-Square | DF | Pr > ChiSq |
| Likelihood Ratio                       | 8.0895     | 3  | 0.0442     |
| Score                                  | 9.3817     | 3  | 0.0246     |

| Testing Global Null Hypothesis: BETA=0 |            |    |            |
|----------------------------------------|------------|----|------------|
| Test                                   | Chi-Square | DF | Pr > ChiSq |
| Wald                                   | 3.2200     | 3  | 0.3589     |

| Analysis of Penalized Maximum Likelihood Estimates |    |          |                |                 |            |          |
|----------------------------------------------------|----|----------|----------------|-----------------|------------|----------|
| Parameter                                          | DF | Estimate | Standard Error | Wald Chi-Square | Pr > ChiSq | Exp(Est) |
| Intercept                                          | 1  | 2.0238   | 1.7793         | 1.2937          | 0.2554     | 7.567    |
| A3                                                 | 1  | -0.2271  | 0.2309         | 0.9674          | 0.3253     | 0.797    |
| A8                                                 | 1  | -0.2880  | 0.2235         | 1.6608          | 0.1975     | 0.750    |
| Log_A15                                            | 1  | 0.2243   | 0.2031         | 1.2198          | 0.2694     | 1.251    |

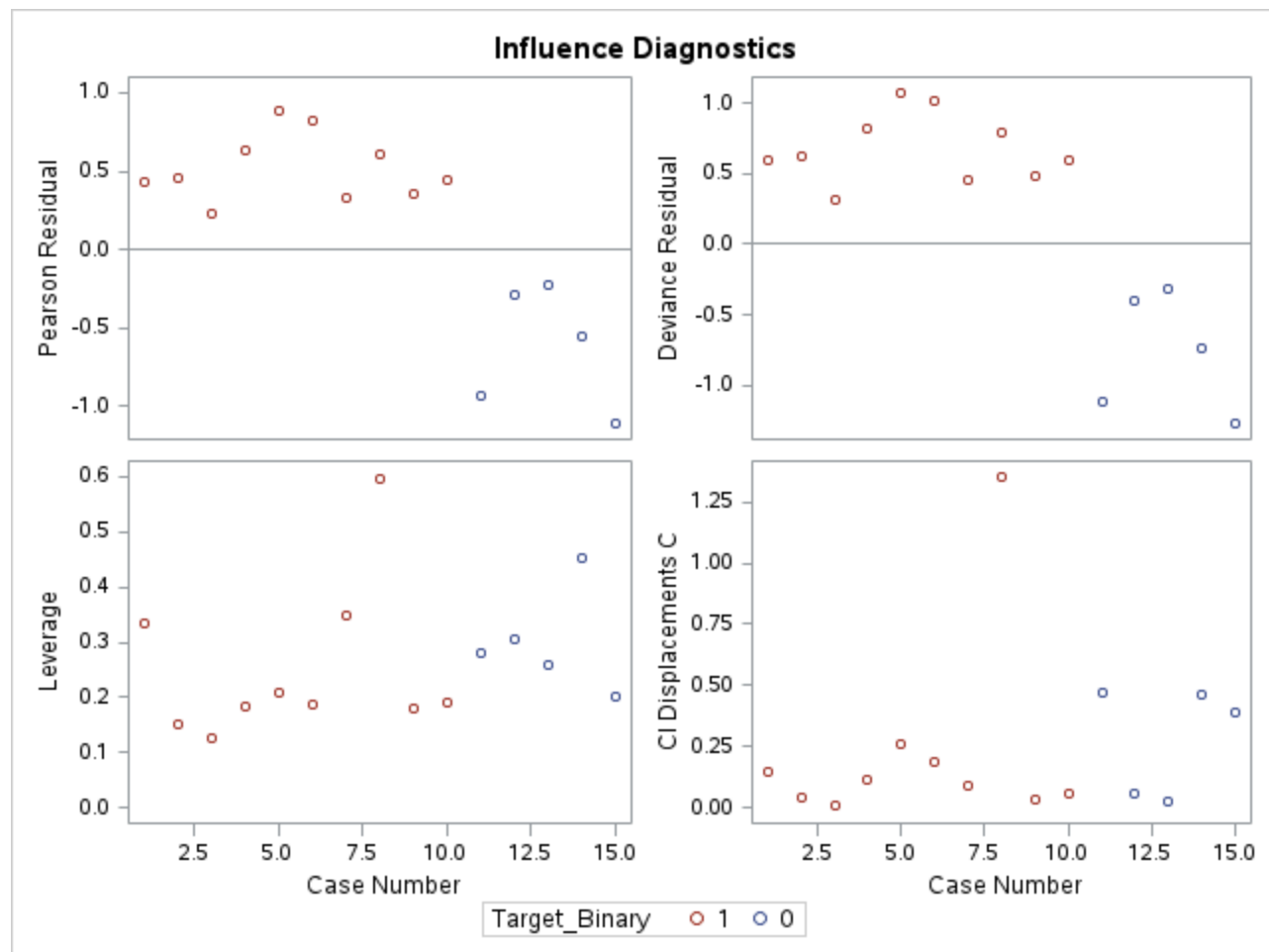
| Odds Ratio Estimates |                |                            |       |
|----------------------|----------------|----------------------------|-------|
| Effect               | Point Estimate | 95% Wald Confidence Limits |       |
| A3                   | 0.797          | 0.507                      | 1.253 |
| A8                   | 0.750          | 0.484                      | 1.162 |
| Log_A15              | 1.251          | 0.840                      | 1.863 |

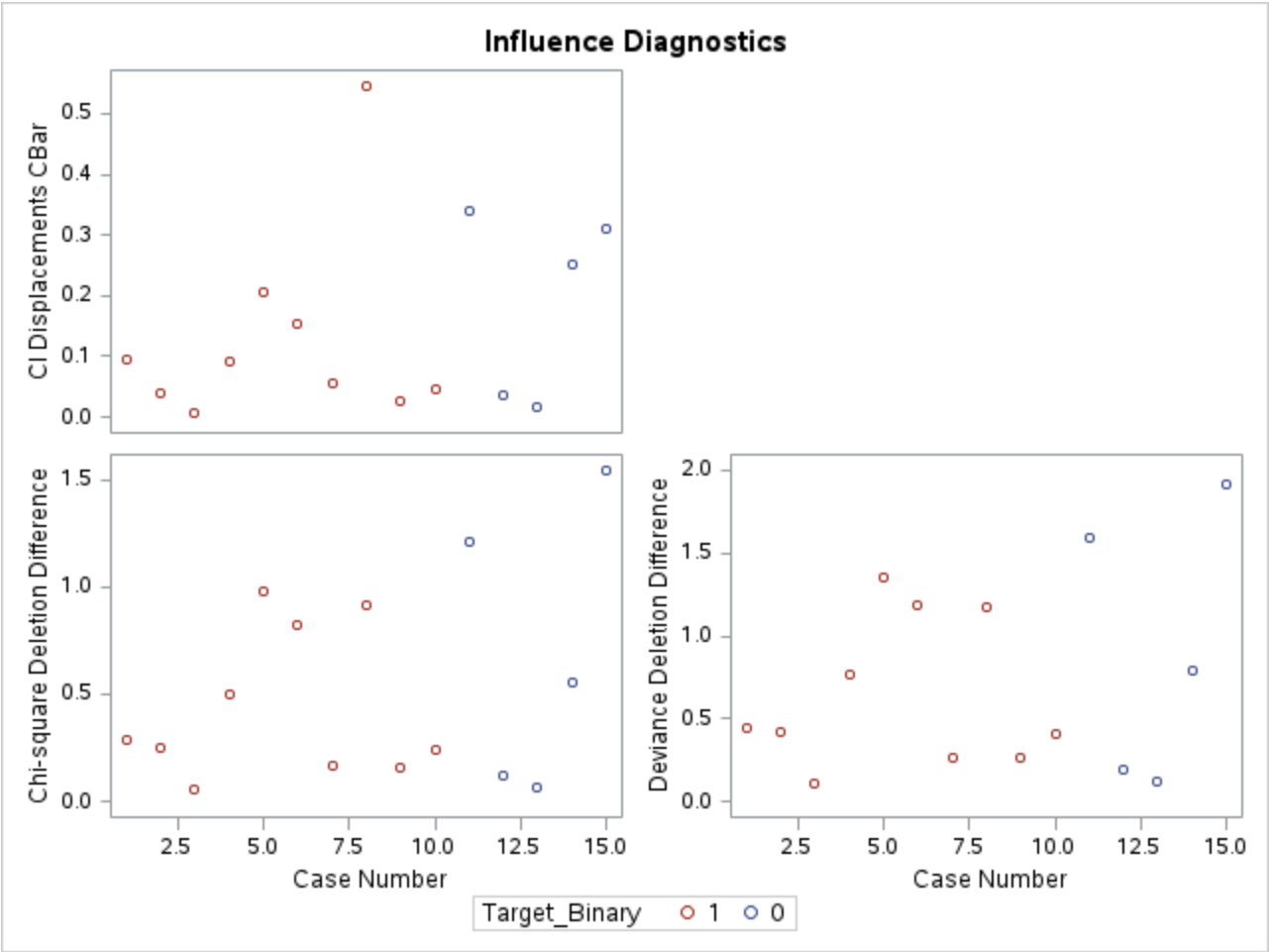
| Association of Predicted Probabilities and Observed Responses |       |           |       |
|---------------------------------------------------------------|-------|-----------|-------|
| Percent Concordant                                            | 100.0 | Somers' D | 1.000 |
| Percent Discordant                                            | 0.0   | Gamma     | 1.000 |
| Percent Tied                                                  | 0.0   | Tau-a     | 0.476 |
| Pairs                                                         | 50    | c         | 1.000 |



| Partition for the Hosmer and Lemeshow Test |       |                   |          |                   |          |
|--------------------------------------------|-------|-------------------|----------|-------------------|----------|
| Group                                      | Total | Target_Binary = 1 |          | Target_Binary = 0 |          |
|                                            |       | Observed          | Expected | Observed          | Expected |
| 1                                          | 2     | 0                 | 0.12     | 2                 | 1.88     |
| 2                                          | 2     | 0                 | 0.70     | 2                 | 1.30     |
| 3                                          | 2     | 1                 | 1.12     | 1                 | 0.88     |
| 4                                          | 2     | 2                 | 1.31     | 0                 | 0.69     |
| 5                                          | 2     | 2                 | 1.56     | 0                 | 0.44     |
| 6                                          | 2     | 2                 | 1.68     | 0                 | 0.32     |
| 7                                          | 2     | 2                 | 1.79     | 0                 | 0.21     |
| 8                                          | 1     | 1                 | 0.95     | 0                 | 0.05     |

| Hosmer and Lemeshow Goodness-of-Fit Test |    |            |
|------------------------------------------|----|------------|
| Chi-Square                               | DF | Pr > ChiSq |
| 3.5318                                   | 6  | 0.7397     |





**Credit Screening Analysis & Risk Engine**

**The HPSPLIT Procedure**

| Performance Information |                |
|-------------------------|----------------|
| Execution Mode          | Single-Machine |
| Number of Threads       | 2              |

| Data Access Information |        |       |           |
|-------------------------|--------|-------|-----------|
| Data                    | Engine | Role  | Path      |
| WORK.UCI_TRANSFORMED    | V9     | Input | On Client |

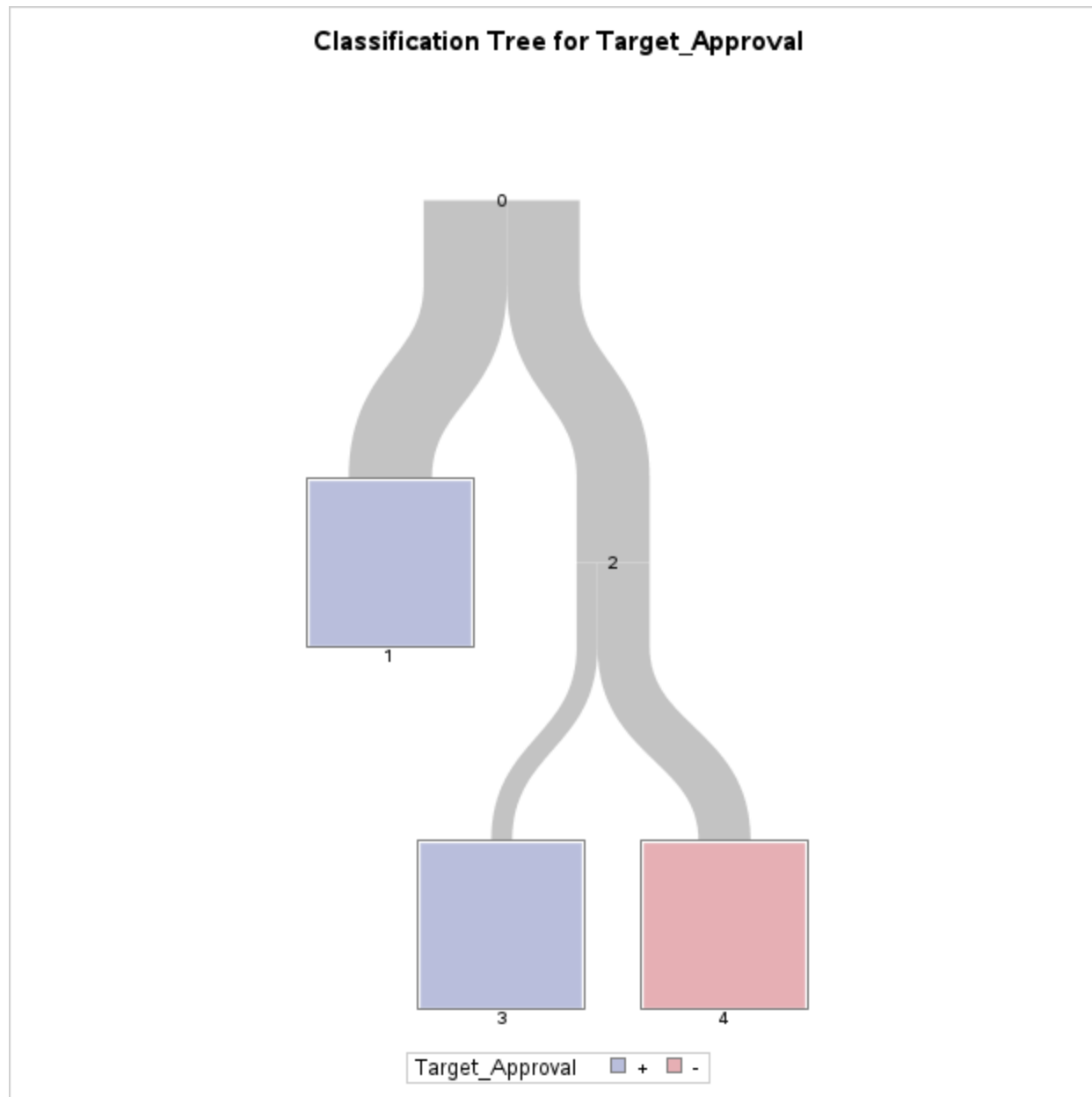
| Model Information               |           |
|---------------------------------|-----------|
| Split Criterion Used            | Entropy   |
| Pruning Method                  | Entropy   |
| Subtree Evaluation Criterion    | Full Tree |
| Number of Branches              | 2         |
| Maximum Tree Depth Requested    | 4         |
| Maximum Tree Depth Achieved     | 2         |
| Tree Depth                      | 2         |
| Number of Leaves Before Pruning | 3         |
| Number of Leaves After Pruning  | 3         |
| Model Event Level               | +         |

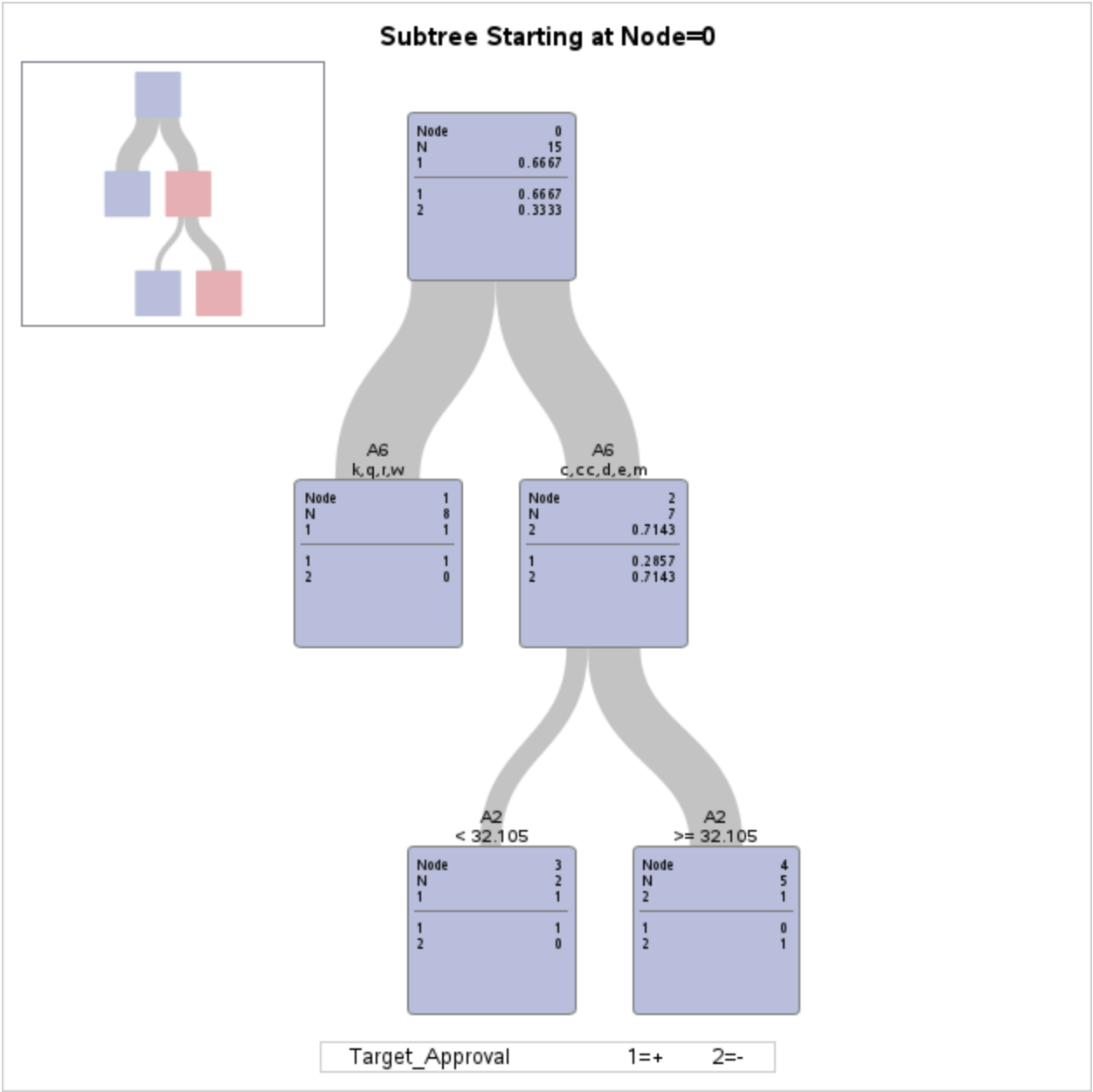
|                             |    |
|-----------------------------|----|
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| Number of Observations Used | 15 |

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Credit Screening Analysis & Risk Engine

The HPSPLIT Procedure



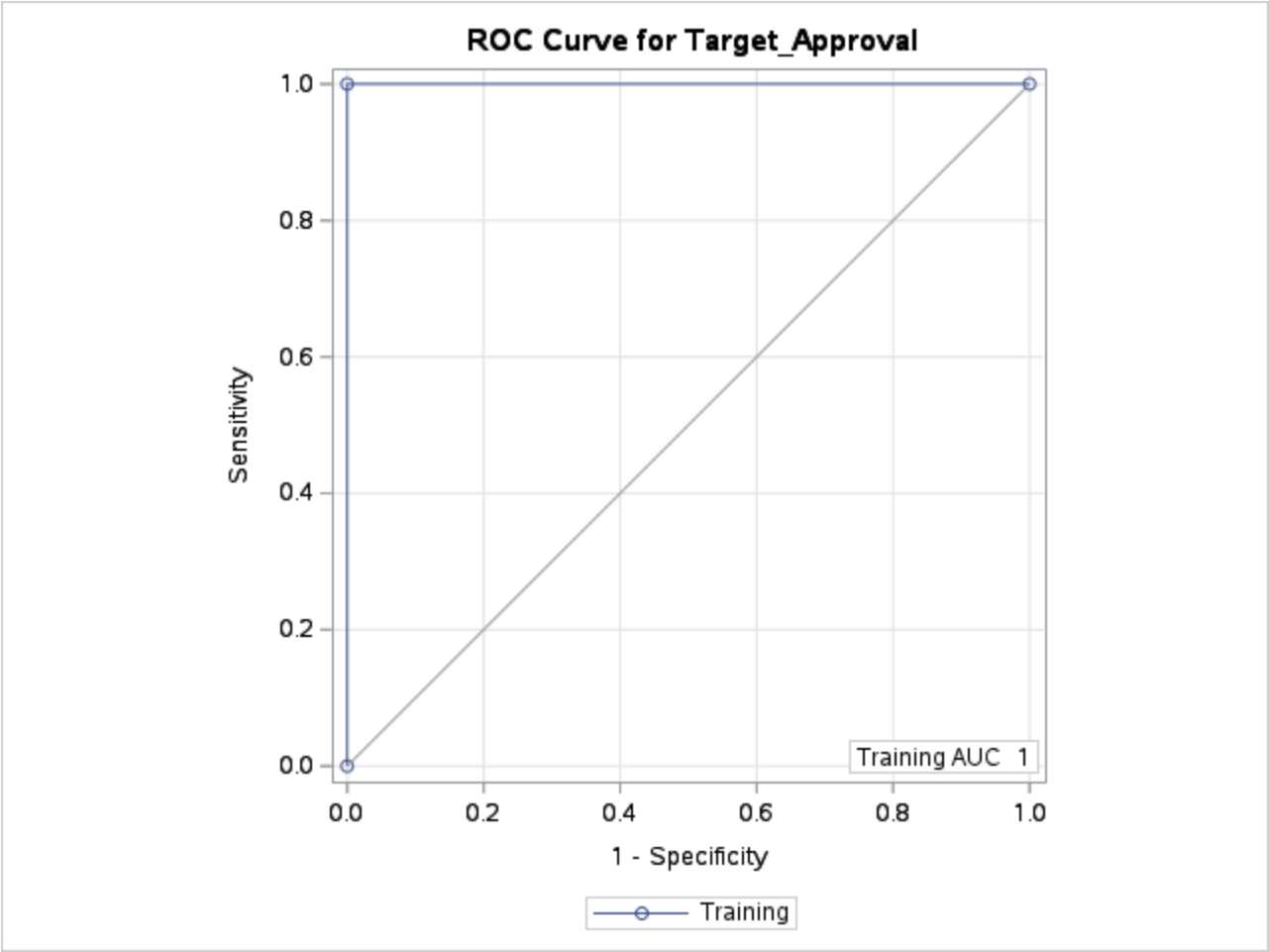


**Credit Screening Analysis & Risk Engine**  
The HPSPLIT Procedure

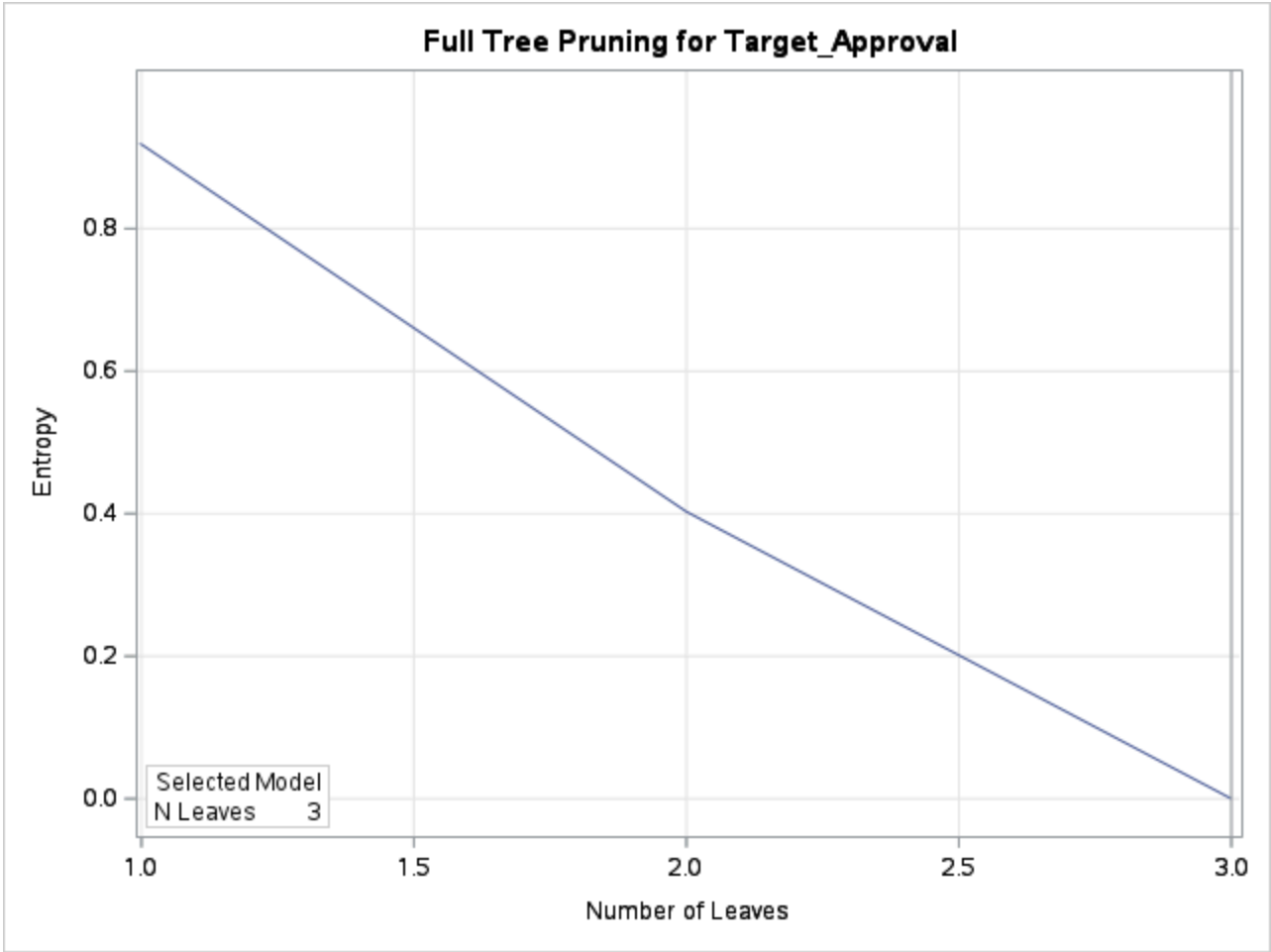


| Model-Based Confusion Matrix |           |   |            |
|------------------------------|-----------|---|------------|
| Actual                       | Predicted |   | Error Rate |
|                              | +         | - |            |
| +                            | 10        | 0 | 0.0000     |
| -                            | 0         | 5 | 0.0000     |

| Model-Based Fit Statistics for Selected Tree |     |           |             |             |         |      |     |        |
|----------------------------------------------|-----|-----------|-------------|-------------|---------|------|-----|--------|
| N Leaves                                     | ASE | Mis-class | Sensitivity | Specificity | Entropy | Gini | RSS | AUC    |
| 3                                            | 0   | 0.0000    | 1.0000      | 1.0000      | 0       | 0    | 0   | 1.0000 |



| Variable Importance |          |            |       |
|---------------------|----------|------------|-------|
| Variable            | Training |            | Count |
|                     | Relative | Importance |       |
| A6                  | 1.0000   | 1.9518     | 1     |
| A2                  | 0.8660   | 1.6903     | 1     |



Decision Distribution for Threshold = 0.45

The FREQ Procedure

|           |
|-----------|
| Frequency |
| Percent   |
| Row Pct   |

| Col Pct | Table of Final_Decision by Credit_Score_Band |                   |          |        |
|---------|----------------------------------------------|-------------------|----------|--------|
|         | Final_Decision                               | Credit_Score_Band |          |        |
|         |                                              | High Ris          | Low Risk | Total  |
|         | APPROVED                                     | 0                 | 12       | 12     |
|         |                                              | 0.00              | 80.00    | 80.00  |
|         |                                              | 0.00              | 100.00   |        |
|         |                                              | 0.00              | 100.00   |        |
|         | REJECTED                                     | 3                 | 0        | 3      |
|         |                                              | 20.00             | 0.00     | 20.00  |
|         |                                              | 100.00            | 0.00     |        |
|         |                                              | 100.00            | 0.00     |        |
|         | Total                                        | 3                 | 12       | 15     |
|         |                                              | 20.00             | 80.00    | 100.00 |

Statistics for Table of Final\_Decision by Credit\_Score\_Band

| Statistic                                                                                       | DF | Value   | Prob   |
|-------------------------------------------------------------------------------------------------|----|---------|--------|
| Chi-Square                                                                                      | 1  | 15.0000 | 0.0001 |
| Likelihood Ratio Chi-Square                                                                     | 1  | 15.0121 | 0.0001 |
| Continuity Adj. Chi-Square                                                                      | 1  | 9.4010  | 0.0022 |
| Mantel-Haenszel Chi-Square                                                                      | 1  | 14.0000 | 0.0002 |
| Phi Coefficient                                                                                 |    | -1.0000 |        |
| Contingency Coefficient                                                                         |    | 0.7071  |        |
| Cramer's V                                                                                      |    | -1.0000 |        |
| WARNING: 75% of the cells have expected counts less than 5. Chi-Square may not be a valid test. |    |         |        |

| Fisher's Exact Test      |        |
|--------------------------|--------|
| Cell (1,1) Frequency (F) | 0      |
| Left-sided Pr <= F       | 0.0022 |
| Right-sided Pr >= F      | 1.0000 |
|                          |        |
| Table Probability (P)    | 0.0022 |
| Two-sided Pr <= P        | 0.0022 |

Sample Size = 15

Expert System Rule Evaluation Summary

|              |        |          |          |             |      |            |         |                  |
|--------------|--------|----------|----------|-------------|------|------------|---------|------------------|
| Applicant ID | Gender | Mean Age | Jobless? | Region Type | Item | Total Loan | Deposit | Rule Reject Flag |
|--------------|--------|----------|----------|-------------|------|------------|---------|------------------|

| Applicant ID | Gender | Mean Age | Jobless? | Region Type | Item      | Total Loan | Deposit | Rule Reject Flag |
|--------------|--------|----------|----------|-------------|-----------|------------|---------|------------------|
| s1           | female | 18       | 0        | normal      | pc        | 30         | 20      | 0                |
| s2           | female | 20       | 0        | normal      | pc        | 40         | 10      | 0                |
| s3           | female | 25       | 1        | problematic | pc        | 48         | 5       | 1                |
| s4           | female | 40       | 0        | normal      | pc        | 84         | 5       | 0                |
| s5           | female | 50       | 0        | problematic | pc        | 48         | 5       | 0                |
| s6           | male   | 18       | 0        | normal      | pc        | 40         | 10      | 0                |
| s7           | male   | 22       | 0        | normal      | pc        | 24         | 10      | 0                |
| s8           | male   | 28       | 0        | normal      | pc        | 40         | 15      | 0                |
| s9           | male   | 40       | 0        | normal      | pc        | 40         | 20      | 0                |
| s10          | male   | 50       | 1        | normal      | pc        | 48         | 5       | 1                |
| s11          | female | 18       | 0        | normal      | car       | 160        | 50      | 0                |
| s12          | female | 20       | 0        | normal      | car       | 200        | 50      | 0                |
| s13          | female | 25       | 0        | normal      | car       | 100        | 50      | 0                |
| s14          | female | 38       | 0        | normal      | car       | 200        | 150     | 0                |
| s15          | female | 50       | 0        | normal      | car       | 300        | 50      | 0                |
| s16          | male   | 19       | 0        | normal      | car       | 140        | 50      | 0                |
| s17          | male   | 21       | 0        | normal      | car       | 60         | 150     | 0                |
| s18          | male   | 25       | 0        | normal      | car       | 200        | 150     | 0                |
| s19          | male   | 38       | 0        | normal      | car       | 200        | 100     | 0                |
| s20          | male   | 50       | 0        | normal      | car       | 300        | 50      | 0                |
| s23          | female | 55       | 1        | problematic | stereo    | 80         | 30      | 1                |
| s30          | female | 68       | 1        | problematic | stereo    | 56         | 2       | 1                |
| s48          | male   | 40       | 0        | problematic | jewel     | 90         | 9       | 1                |
| s56          | female | 43       | 1        | normal      | jewel     | 40         | 10      | 1                |
| s60          | male   | 33       | 1        | problematic | jewel     | 10         | 100     | 1                |
| s68          | male   | 37       | 0        | problematic | medinstru | 500        | 300     | 1                |
| s82          | male   | 67       | 1        | problematic | medinstru | 75         | 400     | 1                |
| s84          | male   | 37       | 0        | problematic | jewel     | 500        | 200     | 0                |
| s85          | female | 52       | 1        | normal      | stereo    | 25         | 100     | 0                |
| s105         | male   | 35       | 0        | problematic | bike      | 100        | 10      | 1                |
| s106         | male   | 18       | 0        | normal      | bike      | 10         | 15      | 0                |
| s115         | female | 25       | 0        | normal      | bike      | 50         | 10      | 0                |
| s116         | female | 40       | 0        | normal      | furniture | 100        | 50      | 0                |
| s125         | male   | 25       | 0        | normal      | furniture | 140        | 20      | 0                |
|              |        | 35       |          |             |           | 4078       | 2211    |                  |